FINAL YEAR PROJECT · CSP650

Cloudie

Secure Cloud Storage with AI Malware Analyzer and AES Encryption



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AGENDA

What I'll cover today

01 Problem & research gap

02 Aim, objectives & scope

03 System architecture & verdict engine

04 AI analyzer, AES encryption & cloud

05 Results & system testing

06 Conclusion & future work

CHAPTER 1 · BACKGROUND

Cloud storage has a security blind spot

As more confidential data moves to the cloud, uploaded files become an attack vector — and conventional scanning often misses what is hidden inside them.

Signature-only

Scanners like VirusTotal only detect malware that already has a known signature.

Zero-day

New and polymorphic malware constantly changes its form to evade detection.

Documents

Trusted files such as PDFs carry embedded scripts and payloads users open without suspicion.

CHAPTER 2 · RESEARCH GAP

Detection and encryption are treated separately

Most systems either scan *or* encrypt. The delay and hand-off between the two create points of security failure. Research also focuses on Android or network traffic — rarely on cloud-native detection for document malware like PDFs.

Cloudie's answer: one workflow where a clean verdict is required *before* a file is ever encrypted and stored.



AIM & OBJECTIVES

A single platform that scans, then secures

To establish a secure cloud storage platform based on AI malware analysis and a supporting AES encryption layer.

Objective 1

Design a secure cloud storage system incorporating a malware analyzer and encryption for sensitive files.

Objective 2

Develop the malware analyzer together with AES-256 encryption to provide layered security for uploads.

Objective 3

Test the system's functionality, detection accuracy, encryption integrity and overall security.

PROJECT SCOPE

Where the project draws its boundaries

PDF documents

Detection is trained and evaluated on PDF files, the project's primary attack vector.

Static analysis

Files are examined structurally without being executed — safe and scalable for the cloud.

AES-256-CBC

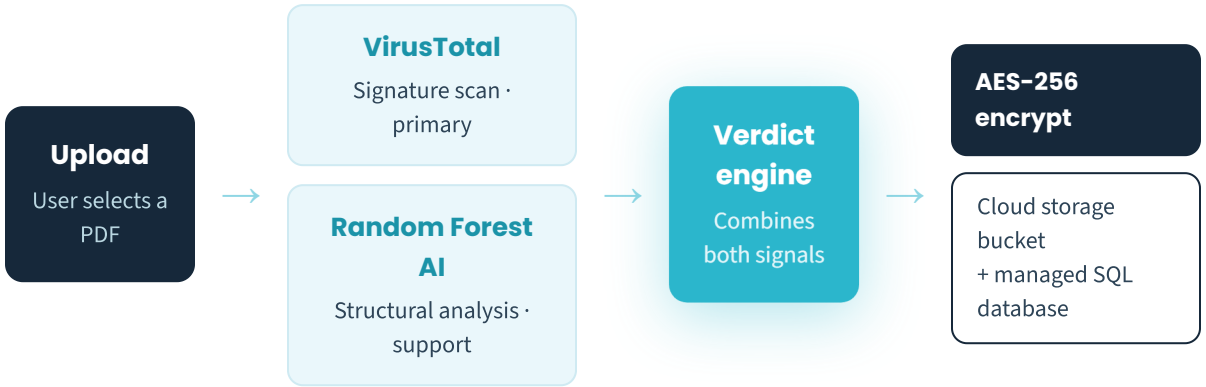
Symmetric encryption chosen for speed and strength on large volumes of cloud data.

Web + cloud

A PHP web app plus a Python Flask ML service, deployed on Google Cloud.

CHAPTER 3 · SYSTEM DESIGN

How a file moves through Cloudie



The encryption key is never stored — only a SHA-256 hash and the initialisation vector are kept for verification.

THE VERDICT ENGINE

Two signals, three outcomes

Malicious

Flagged by VirusTotal — rejected outright, whatever the AI says. Never written to storage.

Suspicious

Clean to VirusTotal but flagged by the AI. Stored, but tagged so the user is aware.

Clean

Both signals agree — the file proceeds to AES-256 encryption and secure storage.

Combining independent detectors removes reliance on a single point of failure — stronger than either method used alone.

AI MALWARE ANALYZER

A Random Forest that reads a file's structure

An ensemble of decision trees votes on each verdict — reducing overfitting and modelling the non-linear patterns that reveal hidden malware, such as embedded JavaScript, self-executing actions and suspicious object structures.

18,126
labelled PDF records

32
structural features

80/20
train / test split

Dataset: CIC-Evasive-PDFMal2022 · scikit-learn Random Forest



ENCRYPTION & KEY HANDLING

The key never lives with the file

Clean files are encrypted with **AES-256-CBC** using a unique key and a random initialisation vector — chosen for strong security with minimal overhead on cloud-scale data.

Shown once

The AES key is delivered as plain text, a QR code and by email — then never again.

Never persisted

Only a SHA-256 hash of the key and the IV are stored for verification.

Separated

Ciphertext sits in the storage bucket; key material in the database — never together.

DEPLOYMENT · GOOGLE CLOUD

Two services, cleanly separated

Cloud Run — PHP web app

User input, file management, AES encryption and server-side logic.

Cloud Run — Python Flask API

Hosts the Random Forest model and coordinates the VirusTotal integration.

Cloud Storage

Encrypted file blobs

Managed MySQL

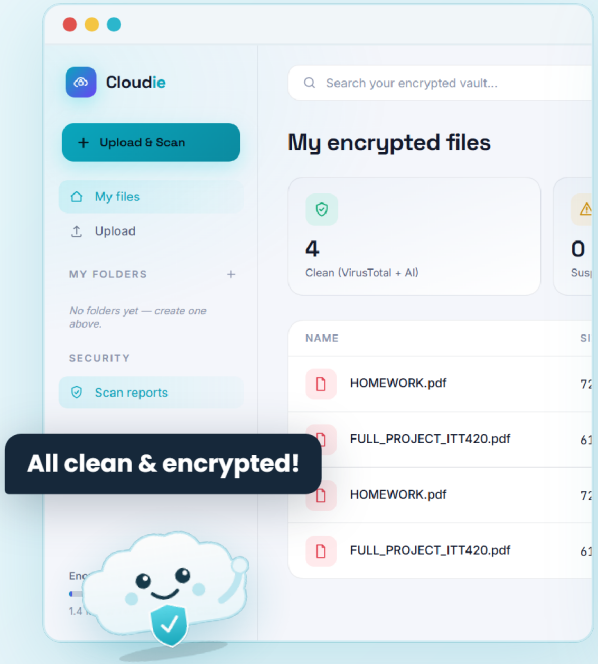
Metadata, verdicts, key hashes



CHAPTER 4 · USER VIEW

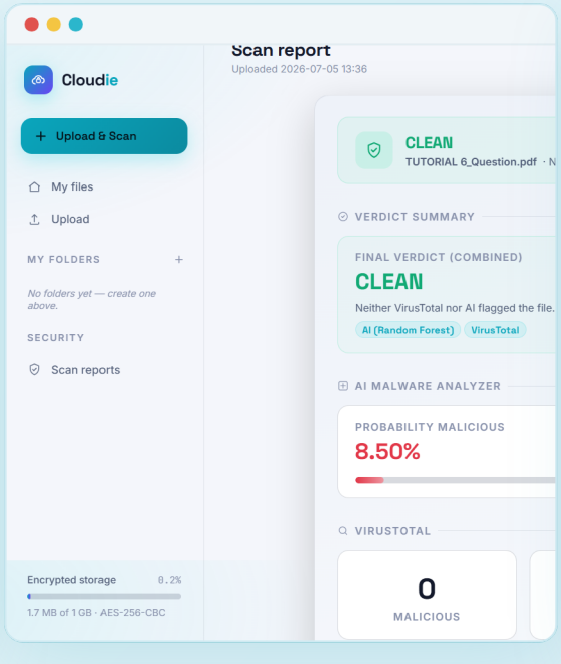
A simple flow over a secure engine

Live from the deployed system



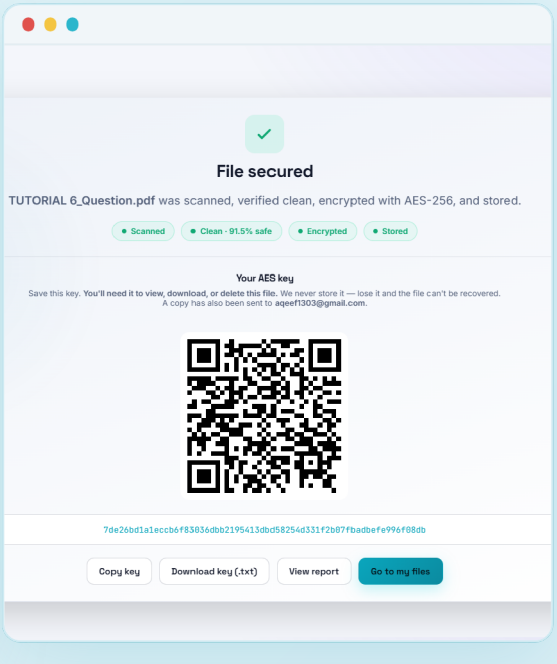
1 Encrypted dashboard

Manage your vault, verdicts at a glance.



2 Combined scan report

VirusTotal + AI verdict, with probabilities.



3 File secured + AES key

Key issued once as text, QR and email.

RESULTS · MODEL PERFORMANCE

Near-perfect detection on the test split

99.86%	100%	99.72%	99.86%
Accuracy	Precision	Recall	F1-score

Precision of 100% means **no benign file was flagged as malicious**; only 5 of 1,813 malicious files evaded detection — high recall is what matters most in a security context.

RESULTS · VALIDATION & ENCRYPTION

Robust detection, negligible overhead

Two validation designs

Design	Accuracy	Recall
Stratified random split	**99.86%**	**99.72%**
Distribution hold-out top quartile by size · n = 4,643	**99.53%**	**97.00%**

Evaluated only on files larger than any seen in training — strong performance is not an artefact of one favourable split.

AES-256-CBC speed

A 10 MB file: **3.5s** encrypt · **3.1s** decrypt. Low and constant across sizes.

Data integrity

Every before/after hash matched (1 KB → 10 MB) — the encryption cycle never alters file content.

RESULTS · SYSTEM TESTING

Every functional test passed

Functional testing — 10 / 10 pass

- ✓ Registration
- ✓ Login
- ✓ File upload
- ✓ Clean-file accept
- ✓ Malicious reject
- ✓ VirusTotal corroboration
- ✓ AES encryption
- ✓ Decryption
- ✓ Download
- ✓ Deletion

Malware corroboration

Clean file	**Accepted**
Known malicious PDF	**Rejected**
VirusTotal-safe file	**Accepted**
Structurally suspicious	**Suspicious**

CHAPTER 5 · REFLECTION

What it delivers — and its limits

Benefits

Strong AES-256-CBC confidentiality for stored files.

Layered detection blocks malware *before* storage.

Combined verdicts are more meaningful than either method alone.

Simple UI and secure email-based recovery — no expertise needed.

Limitations

Detection is specialised to PDF files only.

Model trained on a single dataset (CIC-Evasive-PDFMal2022).

Scanning, encryption and decryption add computational cost.

Cloud Run deployment not yet hardened for production load.

CONCLUSION & FUTURE WORK

All three objectives achieved

Cloudie makes a clean verdict a prerequisite to encryption and storage — a layered, defence-in-depth approach that bridges academic research and a deployable system.

More file types

Extend to Office documents and executables.

More datasets

Validate generalisation beyond one dataset.

Harden deploy

Load testing, auto-scaling, multi-region.

More crypto

Optional RSA / ECC schemes.

THANK YOU

Questions?

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